

Métodos de validación de sólo presencias

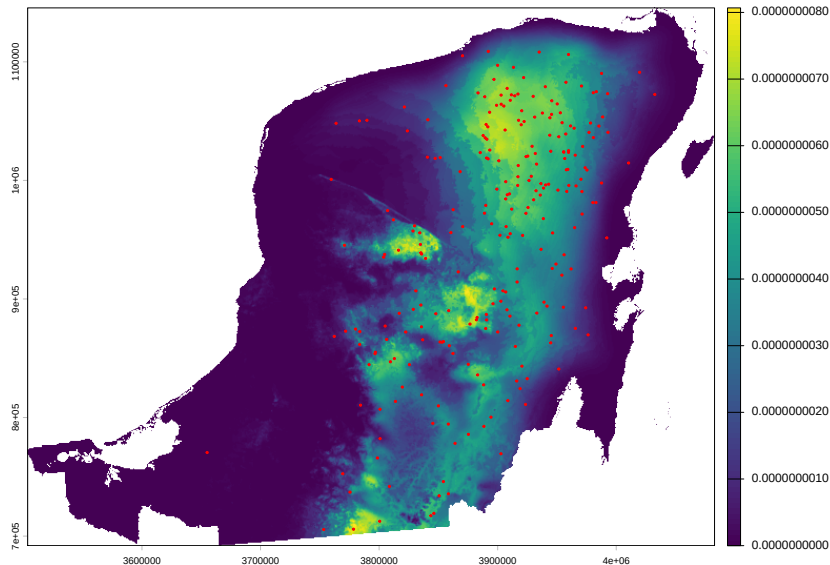
Nombre: Gerardo Martín

2026-08-03

ROC Partial

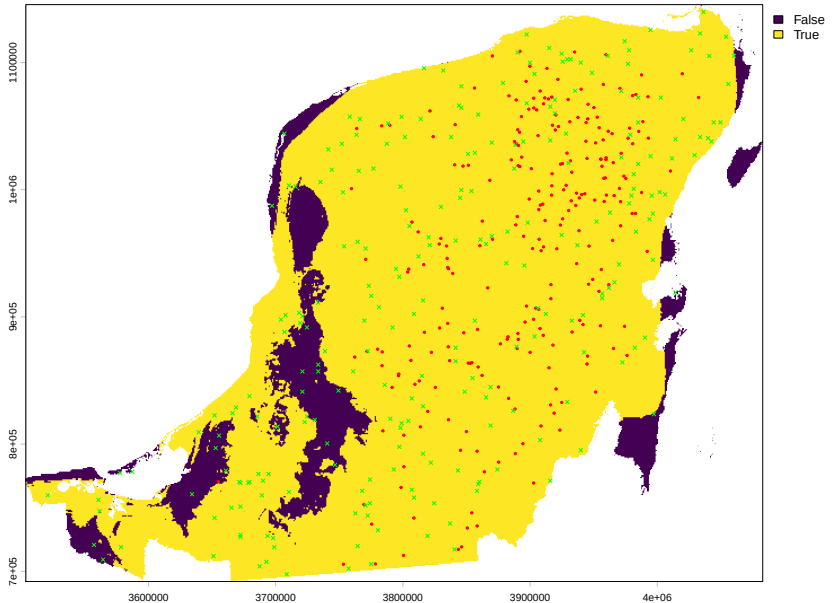
Ejemplo paso a paso

Predicciones de modelo



Umbralización recursiva 1

Umbralización recursiva 2



Fi 0.6 1.1 1.6 2.1 2.6 3.1 3.6 4.1 4.6 5.1 5.6 6.1 6.6 7.1 7.6 8.1 8.6 9.1 9.6 10.1 10.6 11.1 11.6 12.1 12.6 13.1 13.6 14.1 14.6 15.1 15.6 16.1 16.6 17.1 17.6 18.1 18.6 19.1 19.6 20.1 20.6 21.1 21.6 22.1 22.6 23.1 23.6 24.1 24.6 25.1 25.6 26.1 26.6 27.1 27.6 28.1 28.6 29.1 29.6 30.1 30.6 31.1 31.6 32.1 32.6 33.1 33.6 34.1 34.6 35.1 35.6 36.1 36.6 37.1 37.6 38.1 38.6 39.1 39.6 40.1 40.6 41.1 41.6 42.1 42.6 43.1 43.6 44.1 44.6 45.1 45.6 46.1 46.6 47.1 47.6 48.1 48.6 49.1 49.6 50.1 50.6 51.1 51.6 52.1 52.6 53.1 53.6 54.1 54.6 55.1 55.6 56.1 56.6 57.1 57.6 58.1 58.6 59.1 59.6 60.1 60.6 61.1 61.6 62.1 62.6 63.1 63.6 64.1 64.6 65.1 65.6 66.1 66.6 67.1 67.6 68.1 68.6 69.1 69.6 70.1 70.6 71.1 71.6 72.1 72.6 73.1 73.6 74.1 74.6 75.1 75.6 76.1 76.6 77.1 77.6 78.1 78.6 79.1 79.6 80.1 80.6 81.1 81.6 82.1 82.6 83.1 83.6 84.1 84.6 85.1 85.6 86.1 86.6 87.1 87.6 88.1 88.6 89.1 89.6 90.1 90.6 91.1 91.6 92.1 92.6 93.1 93.6 94.1 94.6 95.1 95.6 96.1 96.6 97.1 97.6 98.1 98.6 99.1 99.6 100.1 100.6 101.1 101.6 102.1 102.6 103.1 103.6 104.1 104.6 105.1 105.6 106.1 106.6 107.1 107.6 108.1 108.6 109.1 109.6 110.1 110.6 111.1 111.6 112.1 112.6 113.1 113.6 114.1 114.6 115.1 115.6 116.1 116.6 117.1 117.6 118.1 118.6 119.1 119.6 120.1 120.6 121.1 121.6 122.1 122.6 123.1 123.6 124.1 124.6 125.1 125.6 126.1 126.6 127.1 127.6 128.1 128.6 129.1 129.6 130.1 130.6 131.1 131.6 132.1 132.6 133.1 133.6 134.1 134.6 135.1 135.6 136.1 136.6 137.1 137.6 138.1 138.6 139.1 139.6 140.1 140.6 141.1 141.6 142.1 142.6 143.1 143.6 144.1 144.6 145.1 145.6 146.1 146.6 147.1 147.6 148.1 148.6 149.1 149.6 150.1 150.6 151.1 151.6 152.1 152.6 153.1 153.6 154.1 154.6 155.1 155.6 156.1 156.6 157.1 157.6 158.1 158.6 159.1 159.6 160.1 160.6 161.1 161.6 162.1 162.6 163.1 163.6 164.1 164.6 165.1 165.6 166.1 166.6 167.1 167.6 168.1 168.6 169.1 169.6 170.1 170.6 171.1 171.6 172.1 172.6 173.1 173.6 174.1 174.6 175.1 175.6 176.1 176.6 177.1 177.6 178.1 178.6 179.1 179.6 180.1 180.6 181.1 181.6 182.1 182.6 183.1 183.6 184.1 184.6 185.1 185.6 186.1 186.6 187.1 187.6 188.1 188.6 189.1 189.6 190.1 190.6 191.1 191.6 192.1 192.6 193.1 193.6 194.1 194.6 195.1 195.6 196.1 196.6 197.1 197.6 198.1 198.6 199.1 199.6 200.1 200.6 201.1 201.6 202.1 202.6 203.1 203.6 204.1 204.6 205.1 205.6 206.1 206.6 207.1 207.6 208.1 208.6 209.1 209.6 210.1 210.6 211.1 211.6 212.1 212.6 213.1 213.6 214.1 214.6 215.1 215.6 216.1 216.6 217.1 217.6 218.1 218.6 219.1 219.6 220.1 220.6 221.1 221.6 222.1 222.6 223.1 223.6 224.1 224.6 225.1 225.6 226.1 226.6 227.1 227.6 228.1 228.6 229.1 229.6 230.1 230.6 231.1 231.6 232.1 232.6 233.1 233.6 234.1 234.6 235.1 235.6 236.1 236.6 237.1 237.6 238.1 238.6 239.1 239.6 240.1 240.6 241.1 241.6 242.1 242.6 243.1 243.6 244.1 244.6 245.1 245.6 246.1 246.6 247.1 247.6 248.1 248.6 249.1 249.6 250.1 250.6 251.1 251.6 252.1 252.6 253.1 253.6 254.1 254.6 255.1 255.6 256.1 256.6 257.1 257.6 258.1 258.6 259.1 259.6 260.1 260.6 261.1 261.6 262.1 262.6 263.1 263.6 264.1 264.6 265.1 265.6 266.1 266.6 267.1 267.6 268.1 268.6 269.1 269.6 270.1 270.6 271.1 271.6 272.1 272.6 273.1 273.6 274.1 274.6 275.1 275.6 276.1 276.6 277.1 277.6 278.1 278.6 279.1 279.6 280.1 280.6 281.1 281.6 282.1 282.6 283.1 283.6 284.1 284.6 285.1 285.6 286.1 286.6 287.1 287.6 288.1 288.6 289.1 289.6 290.1 290.6 291.1 291.6 292.1 292.6 293.1 293.6 294.1 294.6 295.1 295.6 296.1 296.6 297.1 297.6 298.1 298.6 299.1 299.6 300.1 300.6 301.1 301.6 302.1 302.6 303.1 303.6 304.1 304.6 305.1 305.6 306.1 306.6 307.1 307.6 308.1 308.6 309.1 309.6 310.1 310.6 311.1 311.6 312.1 312.6 313.1 313.6 314.1 314.6 315.1 315.6 316.1 316.6 317.1 317.6 318.1 318.6 319.1 319.6 320.1 320.6 321.1 321.6 322.1 322.6 323.1 323.6 324.1 324.6 325.1 325.6 326.1 326.6 327.1 327.6 328.1 328.6 329.1 329.6 330.1 330.6 331.1 331.6 332.1 332.6 333.1 333.6 334.1 334.6 335.1 335.6 336.1 336.6 337.1 337.6 338.1 338.6 339.1 339.6 340.1 340.6 341.1 341.6 342.1 342.6 343.1 343.6 344.1 344.6 345.1 345.6 346.1 346.6 347.1 347.6 348.1 348.6 349.1 349.6 350.1 350.6 351.1 351.6 352.1 352.6 353.1 353.6 354.1 354.6 355.1 355.6 356.1 356.6 357.1 357.6 358.1 358.6 359.1 35

Umbralización recursiva 3

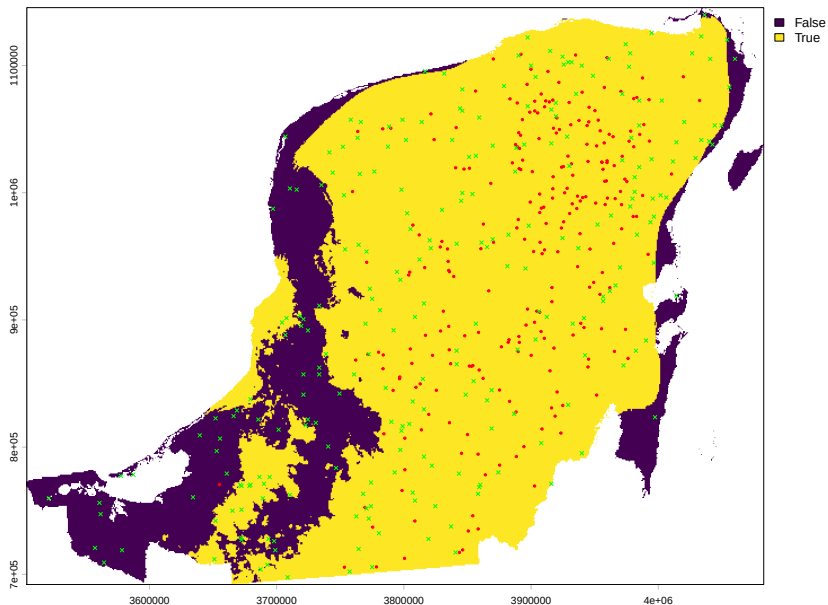


Fig. 2.6. Result of recursive thresholding with a threshold of 0.60.

Umbralización recursiva 5

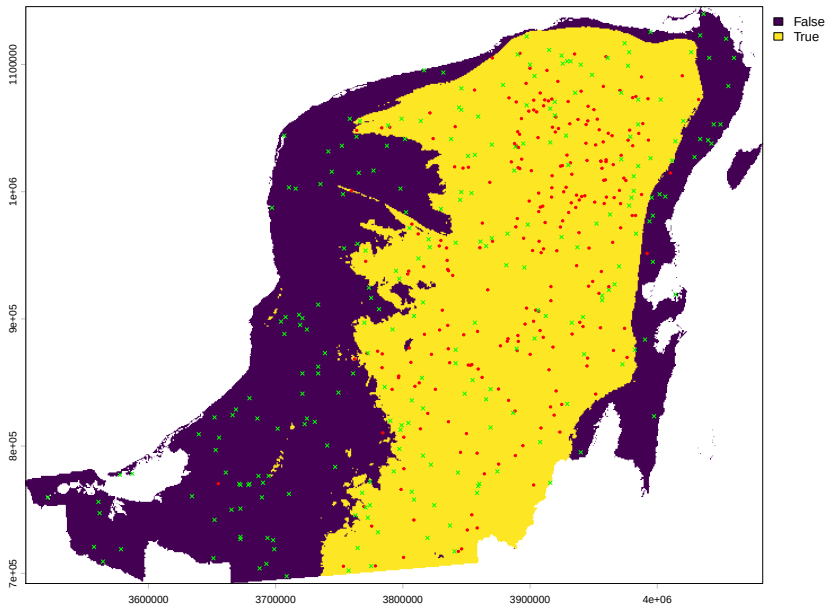
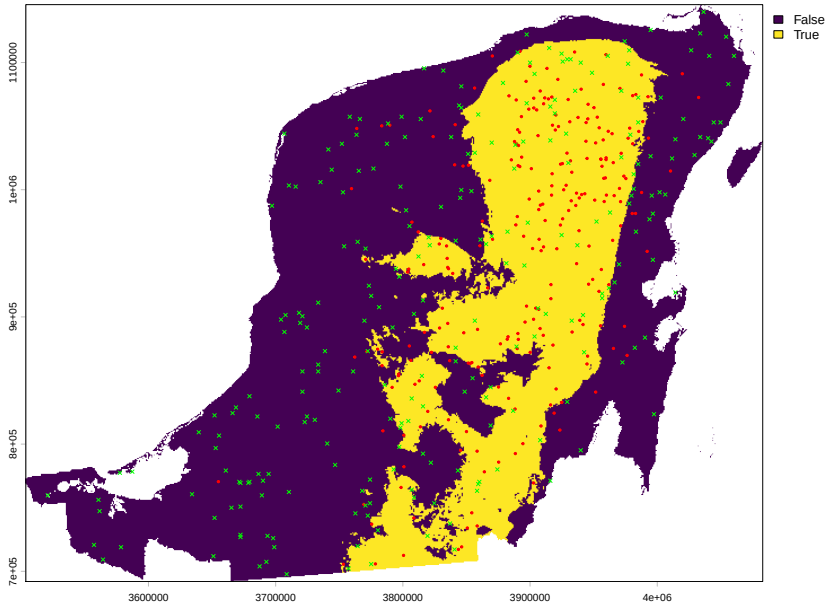


Fig. 4.6. Resultado de la función de umbralización recursiva con $\alpha = 0.004$.

Umbralización recursiva 7


$$F_0 = F_C + \frac{1}{2} F_L + \frac{1}{6} F_{\text{L}}^2 + \dots$$

Umbralización recursiva 9

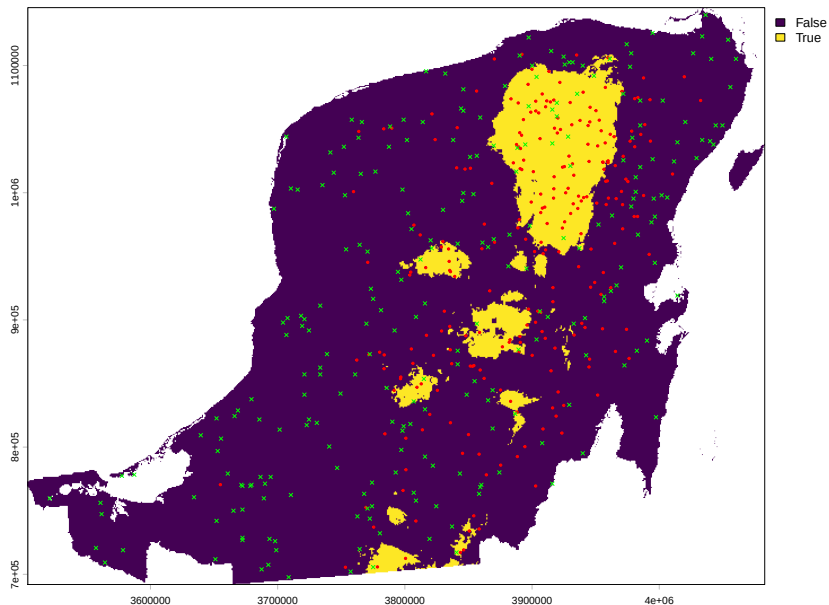


Fig. 6.6. Resultado de la función de umbralización recursiva para el 10.49% de

Umbralización recursiva 10

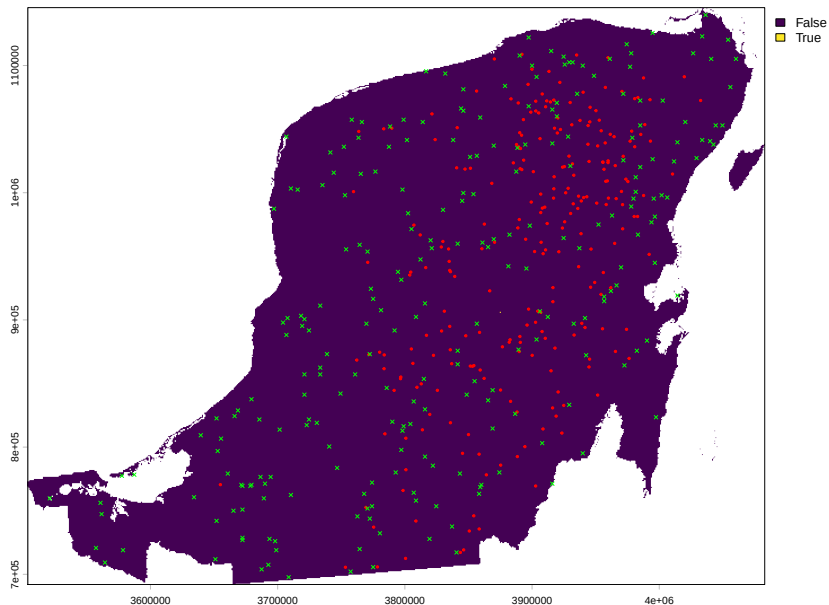
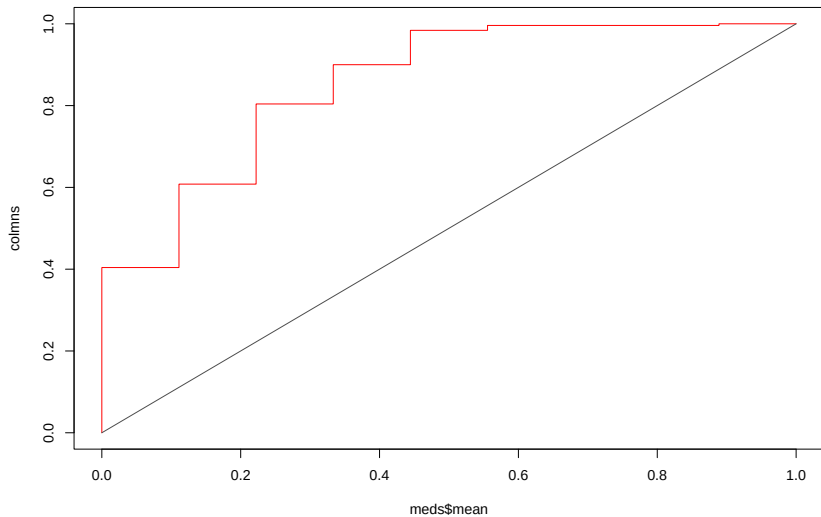


Fig. 3.6. Umbralización recursiva de la imagen de la Fig. 3.5 con un umbral de 10.

La curva ROC



La implementación en espatsmo

La implementación en espatsmo

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## [1] 0
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## [1] 1.5
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Table 1: Simulaciones 1-6 de la roc parcial.

iteration	Area.pres	Area.rand	PartialROC
1	0.814664	0.533088	1.528198
2	0.805240	0.535408	1.503974
3	0.827072	0.574352	1.440009
4	0.814168	0.561008	1.451259
5	0.805016	0.529752	1.519609
6	0.808224	0.555864	1.453996

Gráfica